# From Papers to Progress, with Agents: Governing AI-Assisted Empirical Software Engineering Research

Chapter proposal, *Using AI in Empirical Software Engineering Research* (Springer)

**Intended Authors.** Jason Cusati (PhD student) and Chris Brown (Assistant Professor), Department of Computer Science, Virginia Tech, Blacksburg, VA, USA — djjay@vt.edu, dcbrown@vt.edu.

**Submission Category. Category 2 — Cross-Cutting Topics, Foundations, and Ethics**: methodological rigor, validity and reliability; human-in-the-loop frameworks; reproducibility, transparency and open science; pipeline design. Overlaps Category 1 (artifact and replication-package generation).

**General Goals and Alignment with the Target Audience.** AI is no longer a tool a researcher calls at one step of a study; in empirical software engineering it now plans, retrieves, extracts and critiques, with decisions delegated to it along the way [1]. Current guidance asks authors to disclose models and prompts, keep session traces and validate AI output by hand [2], and stage-specific frameworks now govern literature screening [3] and repository mining [4]. What no standard yet specifies is the research *process* as a governed object: which decisions were delegated to AI and under what authority, which intermediate artifacts they produced, and which human checks accepted them, recorded as reviewable provenance across the whole study. This is the loss of provenance already known to limit knowledge accumulation in software engineering [5], now inside the study and at machine speed—and the same agents can make the remedy affordable.

The chapter operationalizes four published principles for research artifacts (structured, inspectable, long-lived, governed) [5] for AI-assisted empirical research: agents work under bounded authority, claims are recorded as typed entities, decisions leave durable provenance, and reproducibility is built in as executable checks. It is written for PhD students and early-career researchers who already use agents and need to know what to record and where human judgment must stay; for reviewers who need criteria for assessing agent-assisted studies; and for educators, who get a framework and worked examples to assign alongside existing methods guidance. It is grounded in working tooling and two cases from our own practice: an experience report and reference implementation, not a validated method.

**Outline of Chapter Content.**

1. **From AI tool to research participant.** Why agentic AI moves the methodological unit from prompt–output pairs to workflows, and what existing reporting guidance does not cover.
2. **Failure modes of ungoverned AI-assisted research.** Hidden delegated decisions, prompt and model drift, circular validation, non-durable context, schema choices that distort measured results, unrecorded human interventions. Each is tied to its validity threat.
3. **A governance framework for agentic empirical research.** The four principles as design patterns: bounded agent authority with explicit roles; typed claims (existence proof, method claim, general claim) and the baseline each requires; durable provenance of decisions; human semantic gates with stated criteria and placement.
4. **Engineering reproducibility.** Versioned agent definitions and prompts, a control plane separate from research artifacts, executable checks in CI, FAIR packaging.
5. **Two worked cases.** A literature study of agentic software engineering (268 schema-constrained records, separate adversarial verification) and a time-boxed empirical study in which a claims register, pre-registered thresholds and a human re-derivation ledger gated every agent-drafted result—including the lesson that recorded provenance does not govern by itself: a new agent session used the record reliably only when the protocol for loading it was itself governed.
6. **Adoption guide and research agenda.** A minimum governed workflow, a reporting checklist for agent-assisted studies, and open questions: how much process provenance is enough, measuring generator–evaluator independence, model drift in longitudinal studies.

Readers leave with a vocabulary for what agents may decide, a checklist for what to record, and installable templates.

**Intended Supplementary Material.** A FAIR-archived package (Zenodo, MIT/CC-BY): our open-source plugins `agentic-governance` and `agentic-research` (agent role contracts, decision-record, claims-register and verification-ledger templates, conformance-check scripts); the prompts and agent definitions used in both cases; the 219-record, independently verified source dataset behind this proposal (`https://jason.cusati.us/research/governed-agentic-research/`); and a worked example repository showing the full lifecycle.

[1] V. Gomes et al. LLM-assisted empirical software engineering: SLR and research agenda. arXiv:2604.26192, 2026.
[2] S. Baltes et al. Guidelines for empirical studies in SE involving LLMs. arXiv:2508.15503, 2025.
[3] B. Kitchenham et al. Preliminary guidelines for using and evaluating GenAI tools to support SLRs. arXiv:2607.24991, 2026.
[4] V. De Martino et al. A methodological framework for LLM-based mining of software repositories (PRIMES 2.0). arXiv:2508.02233, 2025.
[5] J. Cusati and C. Brown. From papers to progress: rethinking knowledge accumulation in software engineering. ICSE-FoSE 2026, arXiv:2604.16208.